

Assignment 3

Project Overview

In Assignment 2, you built baseline predictors for player value and performance. This final project requires you to evolve those independent scripts into a Unified Scouting System. You are no longer just "training models"; you are designing an optimized, high-stability pipeline that addresses the flaws of your previous iteration (overfitting, variance, and bias).

This project is a direct continuation of your work in [ML_Assignment2.pdf](#). You are tasked with evolving your baseline predictors into a robust, high-stakes system for professional scouting.

1. Justify Your Modeling Choices

In your previous work, you utilized linear assumptions and basic probabilistic models. For this system, you must move beyond these baselines to handle non-linear relationships and complex data patterns.

- **The Model Selection Dilemma:** You must select and implement at least three diverse algorithmic approaches (e.g., instance-based, kernel-based, or tree-based).
- **Decision Rationale:** You must provide a technical justification for why each chosen architecture is appropriate for the FIFA dataset's specific dimensions and distributions.

2. Improve Model Performance Using Appropriate Techniques

A raw model is rarely production-ready. You must demonstrate a systematic approach to extracting maximum performance while maintaining reliability.

- **Systematic Optimization:** You must move beyond default configurations. You are required to use rigorous search methods to find the optimal balance of hyperparameters for every model in your system.
- **Error Diagnosis:** Use validation techniques to prove whether your performance bottlenecks are caused by high bias (underfitting) or high variance (overfitting).
- **Architectural Synergy (Ensembling):** You must design a "Committee of Models." You are responsible for deciding how to combine different learners whether through parallel voting, sequential error-correction, or meta-learning to achieve a result that no single model could reach on its own.

3. Build a System That Predicts an Outcome

Your final deliverable is a unified pipeline that transforms raw player data into actionable insights.

- **Unified Inference:** Construct a single pipeline that accepts a player's profile and simultaneously generates a valuation and a performance tier, ensuring the preprocessing logic is consistent across both tasks.
- **Stability Assessment:** Reliability is paramount. You must provide a statistical proof of your system's stability, demonstrating that its performance remains consistent across different subsets of the data.
- **System Comparison:** You must include a comparative analysis showing exactly how this advanced system outperforms the baseline metrics you reported in [ML_Assignment2.pdf](#).

Submission & Constraints

- **Technical Continuity:** You must maintain the work established in your previous assignment.
- **Ethical Integrity:** Plagiarism and cheating are strictly prohibited and will result in heavy penalties for all involved groups.
- **Project Submission Deadline:** Saturday, June 9th, 2026, at 11:59 PM using the following [form](#).

Deliverables

Each group of **five students** must submit a single compressed folder (GP_XX.zip) containing the following two artifacts:

1. Python Notebook fully commented and executable from top to bottom, must include all visualizations, and contains your work in assignment 2.
2. To facilitate automated cross-group plagiarism checks, you must export a JSON file called results.json with the following keys:
 - `group_members`: List of full names and IDs.
 - `best_hyperparameters`: The final parameters selected via **Grid Search** for each base model.
 - `cv_stability`: The mean and standard deviation of your accuracy/r2 across all **Cross-Validation** folds.
 - **Note:** Identical logs across different groups will trigger an immediate investigation.

Important Note: Team numbers will be assigned and provided as soon as possible.